

Homework 1 (Questions 1 - 6)

Jonathan Morris

Question 1 Population and sample

Conceptual • 8 points

A clinic has 2,000 adults eligible for a sleep program during a specified year. Researchers randomly select 100 of these adults from the complete eligibility list; all participate. After four weeks, 35 report improvement. Everyone receives the same program, and there is no comparison group.

- a. Identify the target population and the sample. (2 points)

Solution: The target population is the 2,000 adults eligible for the sleep program. The sample is the 100 selected individuals.

- b. Define the population response proportion p and the sample statistic $\hat{p}$. Give the observed statistic. (2 points)

Solution: The population response proportion is the proportion of individuals that would respond to the intervention in the population. In contrast, the sample statistic $\hat{p}$ is the proportion of individuals that respond to the intervention in the sample.

In this case:

$$\hat{p} = \frac{35}{100}$$

- c. Classify “35% of these 100 adults improved” and “What proportion of all 2,000 eligible adults would improve under this program?” as descriptive or inferential. (2 points)

Solution:

- “35% of these 100 adults improved” is descriptive.
 - “What proportion of all 2,000 eligible adults would improve under this program?” is inferential.
- d. Does the observed 35% establish that the program caused improvement? Explain what comparison is missing. (2 points)

Solution: No, this proportion does NOT establish improvement because there is no control group. Without a control group, we can’t necessarily conclude whether the 35 percent would have improved without intervention or if the intervention had a direct effect.

Question 2 Study designs and sources of error

Conceptual • 10 points

- a. Name the design in each scenario: (i) survey current shift work and current symptoms once; (ii) use archived records to identify workers free of the disorder in 2020, classify their 2020 exposure, and ascertain new cases through 2025; (iii) select 80 cases and 160 controls, then review earlier exposures; (iv) randomly assign 120 volunteers to counseling or usual care. (4 points)

Solution:

- (i) Cross-sectional
 - (ii) Cohort study
 - (iii) Case-control
 - (iv) Randomized controlled trial
- b. In scenario (iv), distinguish random assignment from random sampling. In scenario (iii), explain why 80/240 is not an estimate of population prevalence. (2 points)

Solution: In scenario (iv), random assignment randomizes whether the participants in the sample get counseling or usual care. Random sampling would have selected the 120 people who participated in the study randomly from the target population. In scenario (iii), we know 80/240 is not an estimate of prevalence because we did not randomly select participants from the population of interest.

- c. Distinguish allocation concealment from blinding by stating when each operates and one problem it addresses. (2 points)

Solution:

Allocation Concealment:

- Begins before the intervention in a trial.
- Conceals the order/randomizes intervention assignment from participants and researchers.
- Addresses biases in researchers choosing which participants get which intervention.

Blinding:

- After/during intervention.
- Blinds people and/or researchers from group assignment during trial.
- Prevents people from leaving the trial if they didn't receive the intervention.

- d. An observational survey measures symptoms differently in the exposed and unexposed groups. Name the type of bias and explain whether increasing the sample size alone fixes it. (2 points)

Solution: This is an example of information bias. No, increasing the sample size does not change the fact that data between the groups is incomparable.

Question 3 Variables and graphical summaries

Conceptual • 8 points

- a. Classify each variable by its scientific meaning: participant ID stored as 01–12; severity recorded as Mild, Moderate, or Severe; age rounded to whole years; number of prior visits; waiting time in minutes; improvement recorded as 0 or 1. Use nominal identifier, ordinal categorical, quantitative discrete, quantitative continuous, or binary categorical as appropriate. (3 points)

Solution:

1. Categorical nominal.
 2. Ordinal.
 3. Quantitative discrete.
 4. Quantitative discrete.
 5. Quantitative continuous.
 6. Binary.
- b. Explain why coding severity as 1, 2, and 3 does not by itself justify treating differences between categories as equal. (1 point)

Solution: Coding severity as mentioned does not establish distance between categories. Specifically, if data are ordinal, then 3 is farther from 1 than 2, but we do not know by how much the distance varies.

- c. Choose a suitable graph for each goal: compare numbers in the Yes and No program categories; display the shape of waiting times; show quartiles and unusually long waits; examine paired age and waiting time measurements. (4 points) **Solution:**

1. Histogram.
2. Histogram or density plot.
3. Boxplot.
4. Scatterplot.

Question 4 Center, spread, and units

Hand calculation • 14 points

The nine observed waiting times, in minutes, are 2, 4, 4, 6, 8, 10, 10, 12, and 25. Treat these as a sample. Use the Lecture 3 linear interpolation convention for quartiles: $h = 1 + (n - 1)p$, with interpolation between the surrounding ordered observations when needed.

- a. Calculate the mean, median, and all modes. (3 points)

Solution:

$$\bar{X} = \frac{1}{9} \sum_{i=1}^9 x_i = \frac{89}{9} = 9$$

Mean = 9, Median = 8, Modes = {4, 10}

- b. Show the deviations from the mean and their squared values. Calculate the sum of squared deviations, sample variance, and sample standard deviation. State their units. (4 points)

Solution:

Deviations from mean:

$$(x_i - \bar{X}) = (-7, -5, -5, -3, -1, 1, 1, 3, 16)$$

Deviations from mean squared:

$$(x_i - \bar{X})^2 = (49, 25, 25, 9, 1, 1, 1, 9, 256)$$

Sum of squared deviations:

$$\sum_{i=1}^9 (x_i - \bar{X})^2 = 376 \text{ minutes}$$

Sample variance:

$$s_X^2 = \frac{1}{9-1} \sum_{i=1}^9 (x_i - \bar{X})^2 = \frac{376}{8} = 47 \text{ minutes}$$

Sample sd:

$$\sqrt{s_X^2} = \sqrt{47} \text{ minutes}$$

- c. Calculate Q_1 , Q_3 , IQR, and the two $1.5 \times \text{IQR}$ fences. Identify any flagged value and the actual lower and upper boxplot whiskers. (4 points)

Solution:

Observe the formula from class:

$$Q(p) = (1 - \gamma)x_j + \gamma x_{j+1}$$

Where

$$h = 1 + (n - 1)p$$

$$j = \lfloor h \rfloor$$

$$\gamma = h - j$$

Hence, we have that $h_1 = 1 + (9 - 1) * 0.25 = 3$, $j = 3$, and $\gamma = 3 - 3 = 0$

and, $h_3 = 1 + (9 - 1) * 0.75 = 7$, $j = 7$, and $\gamma = 7 - 7 = 0$

By the given formulas:

$$Q_1 = x_3 = 4$$

$$Q_3 = x_7 = 10$$

Hence, the IQR is $Q_3 - Q_1 = 10 - 4 = 6$.

Therefore the fences are calculated as:

$$F_1 = Q_1 - (1.5 * IQR) = 4 - (1.5 * 6) = 4 - 9 = -5$$

$$F_2 = Q_3 + (1.5 * IQR) = 10 + (1.5 * 6) = 10 + 9 = 19$$

Where whisker 1 is the lowest value in the data greater than F_1 , and whisker 2 is the greatest value in the data less than F_2 .

Hence, if W_1 and W_2 represent the lower and upper whiskers, respectively, then $W_1 = 2$ and $W_2 = 12$. Based on these fences and whiskers, the only flagged value is 25.

- d. Let $Y = 60X$ convert minutes to seconds. Use the transformation rules to obtain the mean, variance, and SD in seconds. Calculate the sample coefficient of variation and explain whether converting units changes it. (3 points)

Solution:

Observe the vector for Y:

$$Y = (120, 240, 240, 360, 480, 600, 600, 720, 1500)$$

With the proper transformation rules, we don't need to use this vector and can calculate the requested values as follows:

$$\bar{Y} = 60\bar{X} = 60 * 9 = 540 \text{ seconds}$$

$$s_Y^2 = 60^2 s_X^2 = 3600 * 47 = 169,200 \text{ seconds}$$

$$s_Y = 60s_X = 60 * \sqrt{47} \text{ seconds}$$

Observe the following coefficient of variation calculations:

$$CV_X = \frac{s_X}{\bar{X}} * 100\% = \frac{\sqrt{47}}{9} * 100\%$$

$$CV_Y = \frac{s_Y}{\bar{Y}} * 100\% = \frac{60 * \sqrt{47}}{540} * 100\% = \frac{\sqrt{47}}{9} * 100\%$$

Hence, it follows that $CV_X = CV_Y$, so the coefficient of variation does not change when converting units.

Question 5 Joint, conditional, and independent events

Hand calculation • 12 points

Let A denote working the night shift and B denote having current sleep symptoms. The following is a joint probability table for a hypothetical population.

Shift category	Symptoms B	No symptoms B^c	Total
Night shift A	0.18	0.12	0.30
Other shift A^c	0.14	0.56	0.70
Total	0.32	0.68	1.00

- a. Find $P(A)$, $P(B)$, $P(A \cap B)$, $P(A \cup B)$, and $P(A^c)$. Show the addition rule for the union. (4 points)

Solution:

Observe the following results:

- $\mathbb{P}(A) = 0.30$
- $\mathbb{P}(B) = 0.32$
- $\mathbb{P}(A \cap B) = 0.18$
- $\mathbb{P}(A \cup B) = 1 - 0.56$
- $\mathbb{P}(A^c) = 0.70$

Now observe the addition rule for the union:

$$\mathbb{P}(A \cup B) = 1 - 0.56 = 0.44 = 0.30 + 0.32 - 0.18 = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

- b. Calculate $P(B | A)$ and $P(A | B)$. Explain why the denominators differ. (3 points)

Solution:

- $\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \frac{0.18}{0.3}$
- $\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{0.18}{0.32}$

These probabilities differ because we are conditioning on different events. Specifically, because $\mathbb{P}(A) \neq \mathbb{P}(B)$, the formula specifies that $\mathbb{P}(A|B) \neq \mathbb{P}(B|A)$.

- c. Determine whether A and B are independent using the multiplication criterion. (2 points)

Solution:

If A and B are independent, then:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$$

However, observe that:

$$\mathbb{P}(A \cap B) = 0.18 \neq (0.3)(0.32) = \mathbb{P}(A)\mathbb{P}(B)$$

Hence, A and B are not independent.

- d. Are A and A^c mutually exclusive? Are they independent? Support both answers with probabilities. (3 points)

Solution:

Yes, A and A^c are mutually exclusive. Let Ω define the entire sample space.

$$\mathbb{P}(A \cup A^c) = \mathbb{P}(\Omega) = 1 = 0.3 + 0.7 = \mathbb{P}(A) + \mathbb{P}(A^c)$$

Hence, A and A^c are mutually exclusive.

However, A and A^c are not independent. Let $\emptyset$ be the empty set:

$$\mathbb{P}(A \cap A^c) = \mathbb{P}(\emptyset) = 0 \neq (0.3)(0.7) = \mathbb{P}(A)\mathbb{P}(A^c)$$

Therefore, A and A^c are not independent.

Question 6 Total probability and Bayes' theorem

Hand calculation • 12 points

A hypothetical population is partitioned into three nonoverlapping age groups. Each person belongs to exactly one group. G is a specified new health outcome during the next five years.

Age group	Population proportion	$P(G \mid \text{age group})$
Under 50	0.50	0.02
50–64	0.30	0.06
65 or older	0.20	0.12

- a. Use the law of total probability to calculate $P(G)$. Show the weighted terms. (4 points)

Solution: By the law of total probability:

$$\begin{aligned}\mathbb{P}(G) &= \mathbb{P}(< 50)\mathbb{P}(G \mid < 50) + \mathbb{P}(50 - 64)\mathbb{P}(G \mid 50 - 64) + \mathbb{P}(65+)\mathbb{P}(G \mid 65+) \\ &= (0.5)(0.02) + (0.3)(0.06) + (0.2)(0.12) \\ &= 0.01 + 0.018 + 0.024 \\ &= 0.052\end{aligned}$$

- b. Under this model, calculate the expected number with G among 2,500 people. Explain whether this is a guaranteed observed count. (2 points)

Solution:

Based on this model, we would expect $\mathbb{P}(G) * 2500 = (0.052) * 2500 = 130$ to have G . No, this is not a guaranteed count, only an expected count. This 2,500 people only reflects a sample from the larger population and this 130 is the number we would see on average if we sampled 2,500 people repeatedly from the population.

- c. Use Bayes' theorem to calculate the probability that a person is 65 or older given that G occurs. (4 points)

Solution:

By Bayes Theorem observe the following:

$$\begin{aligned}\mathbb{P}(65+ \mid G) &= \frac{\mathbb{P}(65+ \mid G) * \mathbb{P}(65+)}{\mathbb{P}(< 50 \mid G) * \mathbb{P}(< 50) + \mathbb{P}(50 - 64 \mid G) * \mathbb{P}(50 - 64) + \mathbb{P}(65+ \mid G) * \mathbb{P}(65+)} \\ &= \frac{0.024}{0.052} = 0.4615\end{aligned}$$

- d. Explain why the unweighted mean of 0.02, 0.06, and 0.12 is not the population risk for the stated age distribution. (2 points)

Solution: In this case, these unweighted means do not consider that the age distributions have very different sizes within the stated population. Specifically, the unweighted mean assumes each group has an equal share in the population, when really the groups are unequal.

I HAVE COMPILED THE FINAL QUESTIONS IN THE ATTACHED HTML DOCUMENT. SORRY FOR ANY INCONVENIENCE.